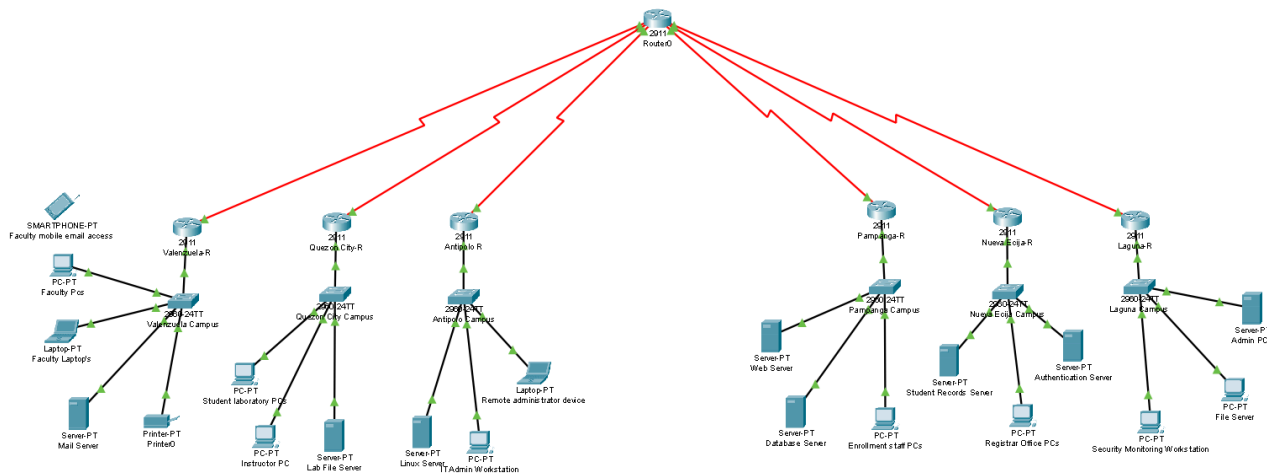




1. The university should prioritize strong security policies such as multi-factor authentication (MFA), regular patch management, and strict access control to protect all six campuses. It should also implement anti-phishing awareness programs to reduce human error. Endpoint protection and antivirus systems must be installed on all devices. Network monitoring should be centralized for faster threat detection. Regular security audits should also be conducted to identify weaknesses early.
2. Zone-Based Policy Firewall (ZPF) improves protection by dividing the network into public, private, and DMZ zones. Each zone has specific security rules that control what traffic is allowed. This prevents unauthorized movement between sensitive and non-sensitive areas. It also limits the impact of attacks by isolating compromised zones. Overall, it creates a more controlled and secure network environment.
3. IPSec VPN is important because it provides secure encrypted communication between campuses. It protects data from interception when transmitted over public networks. It also ensures data integrity so information cannot be altered in transit. Authentication features verify that only trusted campuses can connect. This makes inter-campus communication safe and reliable.
4. If Linux servers in Antipolo Campus are vulnerable, attackers could gain unauthorized remote access. This may lead to data theft or manipulation of sensitive information. Malware or ransomware could be installed on critical systems. Services may be disrupted, affecting students and staff operations. In severe cases, attackers could take full control of the servers.

5. Malware infections from USB devices can be prevented by disabling USB ports on critical systems. The university can also enforce device control policies that only allow approved devices. Endpoint protection software can block malicious files from external drives. Security awareness training can reduce risky user behavior. Regular system monitoring can help detect suspicious activity early.
6. Firewall rules in Laguna Campus should follow a strict “deny all, allow specific” policy. Only necessary ports and services should be opened for communication. Unused or unnecessary traffic should be blocked completely. Outbound traffic should also be monitored to prevent data leaks. Regular updates to firewall rules should be done based on security needs.
7. DDoS attacks on the Pampanga Campus enrollment system can be reduced using traffic filtering techniques. Rate limiting can control excessive requests from suspicious sources. Load balancers can distribute traffic to prevent system overload. Cloud-based DDoS protection services can absorb large-scale attacks. Intrusion prevention systems can detect and block malicious traffic patterns.
8. Student records in Nueva Ecija Campus must be protected using strong password policies. Passwords should be complex and regularly updated. Multi-factor authentication adds another layer of security. This prevents unauthorized access even if passwords are stolen. These measures ensure sensitive data remains protected from attackers.
9. The DMZ serves as a buffer zone between internal networks and external users. It hosts public-facing services like portals and enrollment systems. If attackers compromise the DMZ, the internal network remains protected. It reduces direct exposure of sensitive systems to the internet. This improves overall network security and isolation.
10. Incident response procedures should be standardized across all campuses. A unified response plan ensures consistent handling of security incidents. Clear reporting channels help quickly escalate issues. Trained response teams can react efficiently to threats. Regular drills and reviews improve readiness and coordination.
11. The university should comply with cybersecurity regulations such as data privacy laws. Standards like ISO 27001 help establish proper security frameworks. Compliance ensures proper handling of student and employee data. It also reduces legal and financial risks from breaches. Regular audits help maintain compliance across all campuses.
12. Access Control Lists (ACLs) improve internal security by restricting access between departments. They define which users or systems can access specific resources. This prevents unauthorized lateral movement within the network. ACLs help enforce the principle of least privilege. They also reduce the risk of internal data breaches.

13. Network segmentation separates laboratory systems from administrative offices. This prevents malware from spreading across different departments. It also limits access between critical and non-critical systems. Segmentation reduces the attack surface of the network. It improves overall security and system stability.
14. Monitoring tools like SIEM systems help detect threats in real time. IDS and IPS tools identify and block malicious activities. Centralized dashboards provide visibility across all campuses. Logs and alerts help administrators respond quickly to incidents. Continuous monitoring improves overall cybersecurity response.
15. A redesigned six-zone architecture should use strong network segmentation. ZPF and ACLs should enforce strict traffic control between zones. All campuses should be connected through secure IPSec VPN tunnels. A centralized security system should monitor all activities in real time. This layered approach reduces risks and improves overall resilience.